BAPTISTE PLAQUEVENT-JOURDAIN

PERSONAL INFORMATION

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RESEARCH EXPERIENCE

Postdoc, OPIS-CVN, Gif-sur-Yvette, France *03/2026-02/2027*
Nonsmooth differentiation by conservative mappings
for bilevel optimization with Antonio Silveti-Falls

PhD thesis in international co-supervision *01/2022-08/2025*
List of honor (2026) of the Faculty of Sciences of Université de Sherbrooke
"A Robust Linearization Method for Complementarity Problems
A Detour Through Hyperplane Arrangements": manuscript (in French), slides
With Jean Charles Gilbert, Inria Paris & Jean-Pierre Dussault, Université de Sherbrooke
Nonsmooth analysis/optimization, complementarity, computational geometry

Stage-CDD, Inria Paris, Paris, France *04/2021 – 12/2021*
Master internship and "Pre-thesis" on the PhD's topic

Gap year, Inria Saclay – École Polytechnique, Palaiseau, France *03/2020 – 08/2020*
Internship at RANDOPT: BFGS for multiobjective optimization,
Maximization of the hypervolume of the Pareto front

Gap year, Thales DMS, Élancourt, France *09/2019 – 02/2020*
Internship on modelling of meta-networks for wave redirection

Research Internship, Delft, The Netherlands *05/2019 – 08/2019*
Finite Elements and Newton's method for nonlinear PDEs

STUDIES

Master 2 Optimization, Palaiseau-Orsay – IP Paris *2020 – 2021*
English formation in optimization, analysis, convexity,
optimal control, markov processes, game theory
Classes (September-April) then research internship (Ranked 1st/17)

ENSTA Paris, Palaiseau – general engineer degree *2017 – 2021*
Sciences of Optimization and Data specialization:
Linear optimization, graphs, metaheuristics, complexity
Applied Mathematics option (2nd year)

CPGE (MP), High School Chateaubriand Rennes *2014 – 2017*

ACTIVITIES

Teaching *2021-2024*
Optimization sessions in first and second year at ENSTA and at Sherbrooke
Participation in exercises and exams preparation

MOOC GENIUS for high school students – Polytechnique, Palaiseau *2020 – 2021*
Co-writing of a module of the maths section

PUBLICATIONS

- (published, 2025) *On the B-differential of the componentwise minimum of two affine vectorial functions* (Math. Prog. Comp.) : [DGP25a] (detailed version).
- (online, accepted) *DMS and MultiGLODS: black-box optimization benchmarking of two direct search methods on the bbob-biobj test suite*: [Bro+21].
- (submitted, 2026) *Differentiating Minimal-Norm Solutions to Parametric Optimization Problems*, see preprint [PFS26].
- (submitted, 2026) *Primal and dual approaches for the chamber enumeration of real hyperplane arrangements*: [DGP25b] (detailed version).
- PhD thesis: English manuscript, French manuscript, slides [Pla25] (2-3 other papers in progress).

SOFTWARE

- `ISF.m` : code and documentation of [DGP25a], identification of the chambers of central arrangements and related problems.
- `ISF.jl` : code and documentation of [DGP25b], identification of the chambers of arrangements.

TALKS

- PGMO Days' 26 (parallel session) — November 2026, Palaiseau, France.
- MOP Retreat #4 (group seminar) — October 2026, Saarbruecken, Germany.
- PGMO Days' 25 – slides (parallel session) — November 2025, Palaiseau, France.
- CEA-List DILS seminar – slides (invited by a research team) — October 2025, Palaiseau, France.
- CAS seminar, slides (invited by a research team) — October 2025, Paris, France.
- PhD defense at Inria Paris -- slides — July 2025, Paris, France.
- ISMP 2024, slides (parallel session) — July 2024, Montréal, Canada.
- NOPTA 2024 poster (summer school on nonsmoothness) — April 2024, Anvers, Belgium.
- Julia Days 2023, slides — October 2023, Paris, France.
- Québec colloquium of the ISM – slides (parallel session) — June 2023, Sherbrooke, Canada.
- JOPT2023 – slides (parallel session) — May 2023, Montréal, Canada.
- Annual days of reseach group MOA – slides — October 2022, Nice, France.
- JOPT2022 – slides (parallel session) — May 2022, Montréal, Canada.

GRANTS

ISM Excellence Grant (2000€)– Institute of Mathematical Sciences of Canada *2024*

Globalink Research Grant (4000€) – MITACS-Inria *2023*

Sophie Germain Academic Excellence Grant (master) (10000€) – FMJH *2020 – 2021*

TRIVIA

Software: Julia, Matlab
Hobbies: cooking, board/card/video games

French: mother tongue *English*: fluent (for work)

References

[Bro+21] Dimo Brockhoff et al. "DMS and MultiGLODS: black-box optimization benchmarking of two direct search methods on the bbob-biobj test suite". In: Gecco'21. July 2021, pp. 1251–1258. DOI: 10.1145/3449726.3463207.

[DGP25a] J.-P. Dussault, J.Ch. Gilbert, and B. Plaquevent-Jourdain. *On the B-differential of the componentwise minimum of two affine vectorial functions*. Research Report. 2025.

[DGP25b] Jean-Pierre Dussault, J.Ch. Gilbert, and B. Plaquevent-Jourdain. *Primal and dual approaches for the chamber enumeration of real hyperplane arrangements*. Research Report (in preparation). 2025.

[PFS26] Baptiste Plaquevent-Jourdain, Jalal Fadili, and Antonio Silveti-Falls. *Differentiating Minimal-Norm Solutions to Parametric Optimization Problems*. Research Report. 2026.

[Pla25] Baptiste Plaquevent-Jourdain. "A Robust Linearization Method for Complementarity Problems. A Detour Through Hyperplane Arrangements". PhD thesis. Sorbonne Université et Université de Sherbrooke, 2025. URL: https://theses.hal.science/tel-05253903.